

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 5

1 Computer Networking (awm22)

Topic 4 TCP  
congestion  
control

Recall the simplified TCP throughput equation

$$\text{TCP throughput} = \frac{1.22 \cdot \text{MSS}}{\text{RTT} \cdot \sqrt{L}}$$

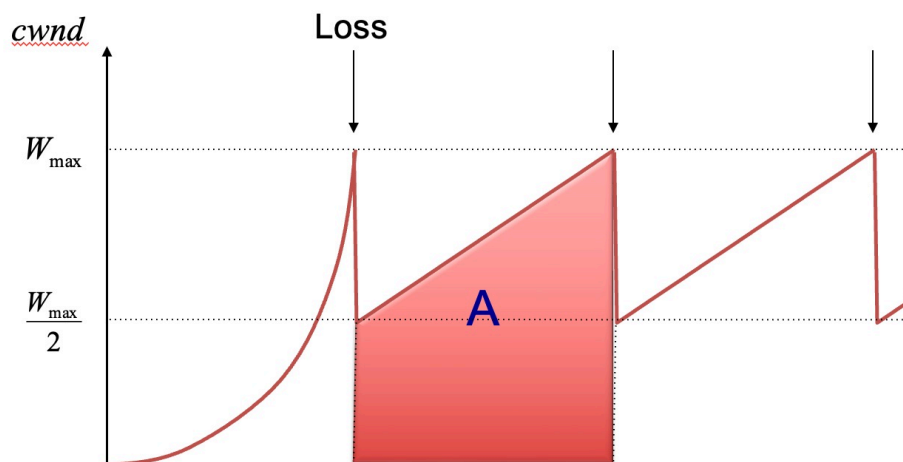
- (a) Provide a derivation of this equation utilising a figure. Include a description of each term of this equation and example values (including units). [6 marks]

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*Answer:*

A derivation is provided in the slides (and discussed in lectures)

## A Simple Model for TCP Thro



Packet drop rate,  $p = 1 / A$ , where  $A = \frac{3}{8} W_{\max}^2$

$$\text{Throughput, } B = \frac{A}{\left(\frac{W_{\max}}{2}\right) \text{RTT}} = \sqrt{\frac{3}{2}} \frac{1}{\text{RTT} \sqrt{p}}$$

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3 marks for derivation + figure

3 marks for definition and examples of each of the following values

MSS is the maximum (TCP) segment size and may be 64kbyte, is by default 576bytes and is no larger than the maximum transmission unit of layer 2 (eg Ethernet).

RTT is the round trip time and ranges from less than 1 millisecond in datacenter networks to 100s of milliseconds for Cambridge-Sydney round trip

L is the loss rate, or occurrence of loss-events (a core feedback in traditional TCP) for very noisy wi-fi it may be 1/100 but is typically much much lower, 1 part in 1 billion and approaching zero for short-lived connections, or those operating through over-provisioned networks

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- (b) What does this equation imply for networks of 10 Gbit/s throughput.

[3 marks]

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*Answer:* For 10 Gbit/s L needs to be (no more than)  $2 \times 10^{-10}$ ; now while such loss rate may be plausible - it implies a huge amount of data needs to be passed to receive this feedback signal. Even with initial slow-start (initial exponentiated growth) this means a single TCP flow will take significant time to achieve full line rate.

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- (c) What important TCP congestion behaviour does this equation not capture?

[3 marks]

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*Answer:* Slow-start. Slow-start is a section of exponentiated growth after the resetting of the window size in a connection. (Such as when a connection starts or when it resumes after a long period idle.) The equation only captures the "saw-tooth" aspect of TCP congestion control as experienced in the congestion-avoidance phase of operation.

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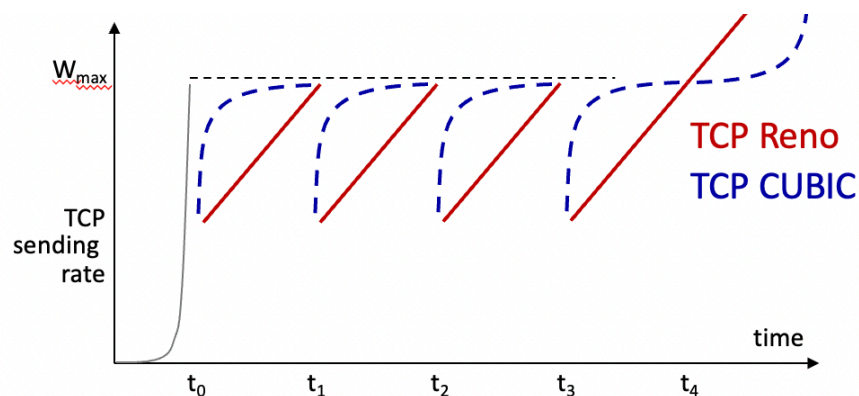
- (d) CUBIC is commonly used as an alternative to classic AIMD TCP congestion control.

- (i) With the aid of a diagram showing window size over time, compare how CUBIC differs from classic AIMD TCP congestion control. [6 marks]

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*Answer:*

I would expect a diagram that resembles those from the notes/slides. in Topic 4 (Transport)



Following loss, CUBIC, instead of a linear increase in window size over time, utilises a CUBIC function that increases window size based upon an estimate to the time the Window max will be met. In this way an assumption of network stability / constant levels of congestion are made and permit the TCP link to increase its window size very quickly, but still slows that increase down as the window gets closer to the last window size that triggered a packet loss event.

Other differences include the decoupling of window size from strict values of RTT - the window grows as a function of time since the last congestion event. This leads to greater fairness among TCP flows of differing RTT.

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- (ii) With reference to the diagram used in part (d)(i) discuss how the CUBIC approach improves performance for a flow on a link with very large bandwidth delay products. [2 marks]

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*Answer:*

By using the CUBIC function, not a linear function, more packets can be maintained in flight due to the larger window size. A diagram of window size over time makes clear a difference between the linear increase and the cubic increase (the area between these two curves represents the improvement CUBIC can make over the linear increase method).

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